

How to use NIST for data security during IoT product development

NIST has created a profile for IoT systems and released a guidelines document [see Reference 7] which mandates the devices' capabilities like networking protocols and cryptographic support to achieve data security.

The usual phases of IoT product development, during which the roles of cryptography and networking protocols are involved, are shown in Figure 2.

Processor selection

Cryptographic algorithms need significant processing capabilities, otherwise it affects performance. The cryptographic operations should be offloaded to hardware accelerators to improve the performance. The selected processor should have this hardware acceleration support.

Firmware selection

The firmware that runs in the constrained nodes should support running cryptographic algorithms and use the hardware acceleration for cryptography. There are some networking protocols that are specifically designed to support constrained nodes. For example, firmware are openthread, contiki, zigbee stack etc.

Application development

During application development, choose a relevant cipher suite and its library implementation to ensure data security.

A cipher suite is a set of various algorithms that help secure a network connection. There are several cipher suites that contain different combinations of algorithms. The kinds of algorithms that cipher suites usually include are listed below:

- Key agreement - To securely exchange keys between devices; e.g., RSA, Diffie-Hellman
- Encryption - To encrypt the data transfer; e.g., AES, DES, RC4

- Message authentication - To provide data integrity checks; e.g., SHA, HMAC
 - Signatures - To authenticate the client or server; e.g., DSA, ECDSA
- To know more about various cipher suites and libraries, check out references 2, 3, 4 and 5 listed at the end of this article. Selection of the relevant cipher suite ensures confidentiality and integrity of the data transfer.

The required cipher suites are generally present as part of the selected firmware. After the cipher suite is selected, the next step is to extract the requirements and best practices related to data security from the guidance documents listed in reference 7, and implement them.

Provisioning of keys

Constrained nodes need preconfigured keys, so that cipher suites use these keys to secure the communication in the data link and transport layers. When the nodes are deployed in the production environment, these keys should be provisioned.

In conventional systems, the focus is more on securing the transport layer, as the medium is wired and the switch/router takes care of the physical security of the data link layer. But in constrained node networks, the focus should be more on the data link layer as the medium is wireless and anybody can access this

medium. Since the security is needed at Layer 2 itself, the only way to achieve this is by using the pre-shared key (psk).

The IoT use cases that require data security are listed below.

1. Network joining – The IoT device joins the network.
2. Data transfer – The IoT device transfers data.
3. Command – The IoT device receives commands from other devices and processes them.
4. Secure storage – The IoT device stores its ID and other data.
5. Network leaving – The IoT device leaves the network.

This article explored the security approach by using the NIST framework which ensures end to end security coverage. The IoT profile generated out of the NIST framework is used as a reference to ensure data security in IoT constrained node networks. The key task is to identify the security requirements from the NIST recommended guidelines. According to Wikipedia, "A 2016 US security framework adoption study reported that in 70 per cent of the surveyed organisations, the NIST cyber security framework is the most popular best practice for information technology (IT) computer security, but many note that it requires significant investment." 

References

- [1] https://en.wikipedia.org/wiki/Cipher_suite
- [2] <https://www.iana.org/assignments/tls-parameters/tls-parameters.xhtml#tls-parameters-4>
- [3] https://en.wikipedia.org/wiki/Comparison_of_TLS_implementations#Protocol_support
- [4] 'Network Security Essentials', 4th Edition (William Stallings)
- [5] Masters' thesis: 'Leveraging Public-key-based Authentication for the Internet of Things' by Hossein Shafagh
- [6] NIST, 'Framework for Improving Critical Infrastructure Cybersecurity', version 1.1
- [7] NIST.IR 8228, 'Considerations for Managing Internet of Things (IoT) Cybersecurity and Privacy Risks'
- [8] <http://blog.cipher.com/a-quick-nist-cybersecurity-framework-summary>
- [9] https://en.wikipedia.org/wiki/Cyber_security_standards
- [10] <https://groups.google.com/forum/#!topic/openthread-users/aLMISCJIWm4>

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